Chapter 6: Causal utility and causal mechanism preservation

6.1 Why causal utility in synthetic data ?

Synthetic data generation optimizes for statistical resemblance while ignoring the processes that generate and measure data.

However, **statistical similarity does not necessarily imply causal similarity**.

For example, two datasets may have similar **means,standard deviations,marginal distributions,correlations**,while having different **covariate combinations**, **treatment assignment mechanisms, outcome generating mechanisms, and treatment effects.**

Therefore, when synthetic data are intended for **causal inference**, we need to ask a stronger question:

> **Does the synthetic dataset preserve the causal structure and causal relationships present in the real dataset?**

Suppose we have a real data of population of 1000 people with covariate age, sex, medication, treatment etc:

And the combination is somewhat :

1. Young + low severe disease - 30%
2. Young + high severe disease - 15%
3.Old + low severe disease=27%
4. Old + high severe disease = 23%

A synthetic dataset may reproduce the **marginal distributions** of age and disease severity reasonably well while generating very different combinations of these variables.

If 900 of 1,000 synthetic observations fall within the estimated support of the real data, then 90% of synthetic observations are supported by the real-data distribution under this particular support definition.

6.2 Causality preservation

Causal utility should not be viewed as a single property. It can be evaluated at **multiple levels**, progressing from preservation of the covariate distribution to preservation of causal effects.

6.2.1 Covariate distribution

The first requirement is that the synthetic population should adequately represent the covariate distribution of the real population

Does the synthetic population resemble the real population?

Possible metrics:

- **α-precision : How much of the synthetic data lies within the distributional support of the real data?**
- **Beta -recall : How much of the real-data distribution is covered by the synthetic data?**

Therefore:
1. High precision + low recall → synthetic data are realistic but may not represent the full population.
2. Low precision + high recall → synthetic data cover the population but may generate unrealistic observations.
3. High precision + high recall → desirable distributional preservation.

6.2.2 Treatment assignment

The question becomes:**Does the synthetic dataset preserve how treatment is assigned conditional on patient characteristics?**

Does the relationship between covariates and treatment remain intact?

- Suppose we have X= covariates, and W = treatment/exposure , we want
  P real(W|X) ≈ P syn(W|X)

Possible evaluation:

- treatment probabilities,
- propensity-score distributions,
- conditional treatment probabilities,
- classification performance for treatment prediction,
- Jensen–Shannon divergence between treatment-assignment distributions.

6.2.3 Outcome mechanism

The question is:**Does the synthetic data preserve the relationship between treatment, covariates, and outcome?**

Does the relationship between treatment, covariates, and outcome remain intact?
Suppose the outcome Y for X= covariates, W = treatment exposure , we want
P real(Y| W,X) ≈ P syn(Y| W,X)

Possible approaches include comparing:

- outcome probabilities,
- regression coefficients,
- predicted outcome distributions,
- conditional outcome distributions,
- predictive performance,
- calibration,
- treatment–outcome interactions.

6.2.4 Causal effect

The key question is:**If we estimate a causal effect using the synthetic data, do we obtain approximately the same causal effect as we would using the real data?**

For a real population: $ATE=E[Y(1)-Y(0)]$. Then we want , **ATE real ≈ ATE synthetic.**

6.3 Measuring the Causal utility

6.3.1 Similar Covariate distribution

Suppose your covariates are: Age, Sex, BMI, Blood pressure, Diabetes
Just comparing the means does not mean the distribution is similar .

What to check

For continuous covariates:

- Mean, SD, median, IQR, quantiles
- density/histogram, KS distance
- Wasserstein distance

For categorical covariates:

- proportions
- frequency tables
- standardized mean differences

And importantly, because these are **covariates**, also examine **joint distributions** such as scatter plots, correlations etc

For example:P(Age,BMI) and P(Age,Diabetes may matter for causal inference.

6.3.1.2 Alpha precision and beta utility:

The Alaa et al. framework defines an **α-support** of the real distribution, S_r^α.

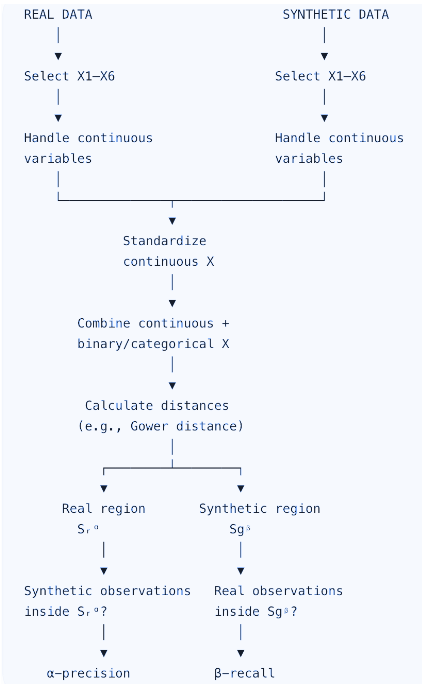
Then: **$P_\alpha=P(X_g \in S_r^\alpha)$** , This is **α-precision**.

In words: Among synthetic observations, what proportion lies within the α-support of the real distribution?

Similarly, define the β-support of the synthetic distribution, S_g^β.

Then: $R_\beta=P(X_r \in S_g^\beta)$, This is **β-recall**.

In words: Among real observations, what proportion lies within the β-support of the synthetic distribution?



6.3.2 Treatment Preservation:

The objective of treatment preservation is to determine whether the synthetic data reproduce the treatment-assignment mechanism observed in the real data.

We want to check the
A) marginal distributions, P real (W=1) vs P syn (W=1), by checking the mean and absolute mean differences in W(treatment /exposure). in real and syn datasets.

B) **Treatment–covariate associations:**
By comparing the treatment proportions across covariate groups; the correlations/associations between W and covariates;etc .

C) **Treatment-assignment model:**
By running the models and checking the coefficients,OR, direction of magnitudes, CI in these models.

D) **Comparing PS and PS score distributions**

6.3.3 Outcome Preservation:

Fit the same model in real and synthetic data.

Compare:

- coefficients
- odds ratios
- confidence intervals
- direction of effects

- predicted probabilities
- calibration
- discrimination

Ref:

1. Amad H, Qian Z, Frauen D, Piskorz J, Feuerriegel S, van der Schaar M. Improving the generation and evaluation of synthetic data for downstream medical causal inference. Advances in Neural Information Processing Systems. 2026 Apr 23;38:30435-78.

2. Xu Y. Generative Synthetic Data for Causal Inference: Pitfalls, Remedies, and Opportunities. arXiv preprint arXiv:2604.23904. 2026 Apr 26.